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A MoE-LLM-based multisensor flexible fusion fault diagnosis method for rotating machinery

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Highlights

- Proposed a MoE-LLM-based multisensor fault diagnosis method for rotating machinery.
- Designed a flexible embedding layer for diverse sensor signal combinations.
- Employed dynamic routing for sparse activation and adaptive feature extraction.
- Introduced curriculum learning to enhance transferability across different scenarios.

Abstract

As a core component of industrial systems, rotating machinery requires accurate fault diagnosis to ensure safety and efficient operation. In recent years, although intelligent diagnostic methods based on Large Language Models (LLMs) have made notable progress, most existing approaches are limited to single-type signals and lack effective modeling of the complementary and synergistic information across multiple sensors, rendering them insufficient for complex and dynamic industrial environments. To address this limitation, this paper proposes a MoE-LLM-based multisensor flexible fusion fault diagnosis method for rotating machinery. The proposed method designs a multisensor embedding layer to map various combinations of sensor signals into unified feature embeddings compatible with LLMs. A sparsely activated Mixture of Experts (MoE) mechanism, comprising both uni-signal and fusion-signal experts, is introduced to enable adaptive modeling and fault identification from multisource signals. Additionally, a curriculum learning-based staged training strategy is developed to enhance the model's

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ducted on three multisensor datasets demonstrate that

the proposed method outperforms mainstream approaches in terms of diagnostic accuracy, robustness, and scalability.

Keywords

Large language model; Mixture of experts; Fault diagnosis; Multisensor fusion; Rotating machinery

1. Introduction

Rotating machinery is extensively used in critical industrial fields such as energy [1], transportation [2], and manufacturing [3], where its condition has a direct impact on the efficiency and safety of production systems [4]. Due to prolonged operation in harsh environments such as high speed and heavy load, failures often occur in key components including bearings, gears, and motors. These failures can lead to production downtime or even safety hazards. Therefore, the developing efficient and reliable fault diagnosis methods is essential for achieving predictive maintenance and maintaining stable operation in industrial systems [5], [6], [7], [8], [9].

In recent years, data-driven intelligent fault diagnosis methods based on deep learning have developed rapidly [10]. By leveraging deep models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformers, researchers have achieved end-to-end signal feature learning and fault recognition, significantly improving diagnostic accuracy [11], [12], [13]. However, most of these methods are designed for single-source signals and lack the ability to effectively exploit complementary information across multiple sensors, which limits their applicability in complex industrial environments.

Regarding these issues, multisensor fusion techniques [14], [15] have been widely investigated. For example, Li et al. [16] proposed a data-level fusion method named Correlation Variance Contribution to integrate vibration signals, while Ye et al. [17] developed a Multisensor Residual Convolution Fusion Network (MRCFN) to capture complementary features from acoustic and vibration signals. In addition, Li et al. [18] introduced MCFormer, a multi-branch cross-fusion Transformer, which leverages parallel perception networks to extract heterogeneous features from different sensors and employs a dynamic classifier to achieve accurate global representations. Similarly, Shi et al. [19] proposed Res-GTLA-Net, a residual network with grouped two-level attention, which avoids feature dilution caused by signal mixing and enhances critical sensor information through intra- and inter-group attention. Despite these advances, most existing approaches remain static and task-specific: they rely on predefined fusion strategies or fixed network structures, lack adaptability to diverse sensor configurations, and are unable to cope with signal absence or heterogeneity. This limits their applicability in real-world industrial scenarios.

Large Language Models (LLMs) [20], [21] provide a promising new paradigm to overcome these challenges. LLMs possess strong contextual understanding, cross-task generalization, and the ability to represent heterogeneous information in a unified semantic space, making them particularly suited for complex diagnostic scenarios. Specifically, they reduce the reliance on large-scale labeled datasets by leveraging pre-trained knowledge, provide a universal interface to align heterogeneous sensor features into token-based representations, and incorporate external domain knowledge to enhance interpretability and robustness, thereby bringing new opportunities for intelligent fault diagnosis [22], [23], [24]. These properties bring new opportunities for intelligent fault diagnosis [22], [23], [24]. Recent studies have begun to explore these advantages by converting sensor signals into textual or embedding inputs [25]. For instance, Zhang et al. [26] transformed observed feature variables into text and fine-tuned LLMs for fault classification, while Tao et al. [27] proposed a framework that converts vibration sequences into LLM-compatible feature vectors and validated its cross-condition performance. However, current LLM-based fail to fully exploit the complementary nature of

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multisensor data, and adopt rigid input representations that cannot flexibly handle signal absence or heterogeneity. Moreover, most approaches rely primarily on text generation tasks and lack the capability to deeply model the structural characteristics of raw physical signals.

To address these limitations, this paper proposes a MoE-LLM-based flexible multisensor fusion fault diagnosis method for rotating machinery. A multisensor embedding layer is designed to flexibly transform various sensor signal combinations into LLM-compatible feature embeddings. A Mixture of Experts (MoE) mechanism is introduced within the Transformer architecture, comprising both uni-signal and fusion-signal expert subnetworks. The MoE mechanism enables sparse activation and adaptive feature extraction through dynamic expert selection, thereby improving diagnostic accuracy and robustness. A curriculum learning-based training strategy is further proposed to divide training into multiple stages of increasing complexity, enabling the model to gradually adapt to more complex tasks and enhancing capabilities. Experimental results on three multisensor datasets demonstrate the effectiveness and superiority of the proposed method. The main contributions of this paper are summarized as follows:

(1) A flexible multisensor embedding layer is designed to accommodate various combinations of sensor inputs. It includes signal preprocessing, feature extraction, and alignment modules that convert multisensor signals into LLM-compatible embeddings.

(2) A novel MoE-LLM architecture is developed, consisting of uni-signal and fusion-signal expert subnetworks. The MoE mechanism enables sparse activation and adaptive feature extraction, enhancing diagnostic performance.

(3) A curriculum learning-based training strategy is introduced to gradually increase task complexity, thereby improving the model's transferability for different scenarios.

The remainder of this paper is organized as follows: Section 2 introduces the background and theoretical foundations. Section 3 presents the overall framework and module designs of the proposed method. Section 4 details the experimental setup and result analysis. Finally, Section 5 concludes the paper and outlines future research directions.

2. Preliminaries

2.1. LLM-based fault diagnosis

LLMs are deep learning-based natural language processing models with strong capabilities in text generation and understanding. In recent years, LLMs have achieved remarkable success in fields such as natural language processing and computer vision [24], [28]. LLMs are typically built on the Transformer architecture [29], consisting of modules such as tokenization, embedding, positional encoding, multi-head self-attention, and feed-forward networks, enabling powerful contextual modeling and representation learning. The training of LLMs generally follows a two-stage pipeline: pre-training and fine-tuning. In the pre-training stage, LLMs adopt either autoregressive or autoencoding structures (e.g., GPT [30] or BERT [31]) to learn statistical patterns and contextual dependencies from large-scale corpora, aiming to minimize language modeling loss such as next-token prediction. In the fine-tuning stage, LLMs are further optimized on task-specific data to enhance performance in downstream tasks such as question answering, classification, and diagnosis.

In recent years, LLM-based fault diagnosis methods have attracted increasing attention. As illustrated in Fig. 1, these methods typically fine-tune pre-trained LLMs such as ChatGPT [32] or LLaMA [33] to adapt to various diagnostic tasks. Depending on the input construction approach, they can be categorized into two types:

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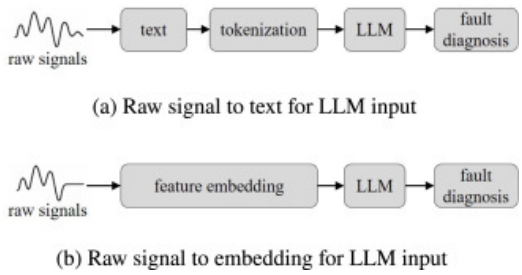

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Fig. 1. Overview of LLM-based fault diagnosis methods.

(1) Signal-to-text conversion: This approach converts raw signals into textual format and inputs them into the LLM for diagnosis. For instance, the raw signal values or extracted features can be directly represented as text [25], [27]. Its advantage lies in directly leveraging the powerful text generation and understanding capabilities of LLMs. However, it may lose important structural information inherent in the raw signal, potentially leading to performance degradation.

(2) Signal embedding representation: This method transforms raw signals into embedding representations, which are then fed into the LLM. For example, signal sequences can be divided into patches with added positional encodings and directly processed by the LLM's Transformer module [23], [27]. This approach preserves the structural information of the signal and better utilizes the contextual modeling power of LLMs. However, it requires carefully designed embedding methods to ensure that the input representations are compatible with the LLM.

Existing LLM-based fault diagnosis methods primarily focus on single-sensor scenarios, utilizing information from only one type of signal while neglecting the complementary and synergistic relationships among multiple signal types. Although some studies have attempted to incorporate multisensor observations by converting them into structured textual inputs, these approaches typically discard the raw signal information, making it difficult to fully exploit the underlying features in complex multisensor data. Moreover, current LLM-based diagnostic frameworks often assume that the types and number of input sensors are fixed. However, in practical industrial applications, due to variations in monitoring conditions and sensor deployment, the input signals across different scenarios are often inconsistent. Such inconsistency significantly limits the adaptability of existing LLM-based approaches to diverse real-world environments. Therefore, there is a critical need for a new method that can effectively address the challenges of multisensor signal fusion and fault diagnosis under varying input conditions.

2.2. MoE mechanism

MoE is a sparsely activated neural network architecture characterized by low computational cost and large model capacity, as only a subset of experts is activated during inference. MoE mechanisms have achieved notable success in natural language processing and computer vision tasks [34], [35]. As illustrated in Fig. 2, the core idea is to employ multiple expert networks to model the input and use a gating network to dynamically select a subset of experts for inference computation.

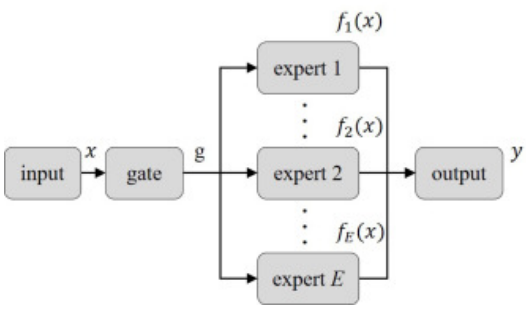

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Fig. 2. Overview of MoE mechanism.

The MoE module typically consists of the following two components:

(1) Expert Networks: Assume there are E experts, each being a sub-network denoted as $f_1, f_2, \ldots, f_E$.

(2) Gating Network: Given an input x, it outputs a gating vector $g \in R^E$ to select experts.

Given an input x, the forward inference process of the MoE module includes:

$$g = \text{Gating}(x) \in R^E \tag{1}$$
$$y = \sum_{i=1}^{E} g_i \cdot f_i(x) \tag{2}$$

where g is the gating vector output by the gating network, and y is the final output of the MoE module. The gating vector g indicates the weights assigned to each expert, and only a subset of experts is activated based on the gating mechanism. The experts are typically trained to specialize in different aspects of the input data, allowing the MoE module to adaptively select the most relevant experts for each input.

However, in LLM-based fault diagnosis tasks, the application of the MoE mechanism remains limited. Existing methods often lack deep understanding and effective feature extraction across multiple signal types. Therefore, embedding signal-specific expert sub-networks into LLMs, and enabling dynamic routing to selectively activate these experts, holds great promise for multisensor signal fusion and fault diagnosis.

3. Methodology

3.1. Overall framework

The overall architecture of the proposed network, including its inputs and outputs, is illustrated in Fig. 3. It comprises four main components: the input prompt, the multisensor embedding layer, the MoE-LLM module, and the output response. The input prompt is constructed from multisensor signals along with their corresponding textual descriptions. The multisensor embedding layer transforms both the raw signals and associated text into feature embeddings compatible with LLMs. The MoE-LLM module performs signal fusion and fault diagnosis through a sparse MoE mechanism. The final output is a diagnostic result along with its corresponding textual explanation. The subsequent sections provide a detailed introduction to each module of the proposed method, as well as the training strategy.

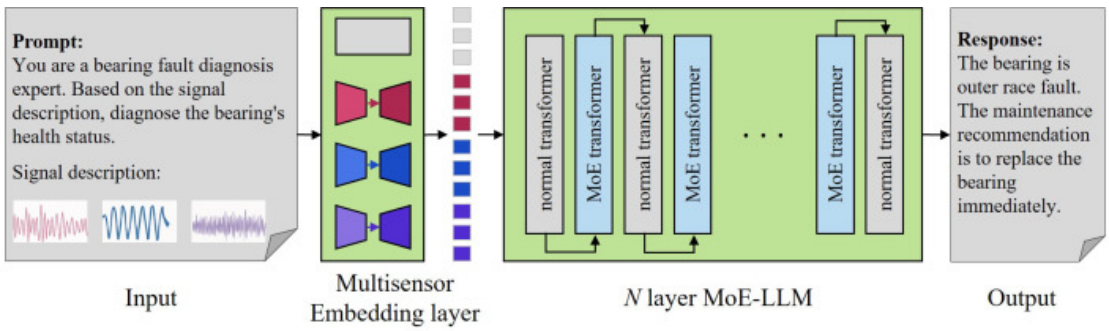

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Fig. 3. The network structure of the proposed method.

3.2. Multisensor embedding layer

The embedding layer in traditional LLMs is primarily designed to process textual data. However, in the context of rotating machinery fault diagnosis, the input data typically consists of multisensor sensor signals. Therefore, it is

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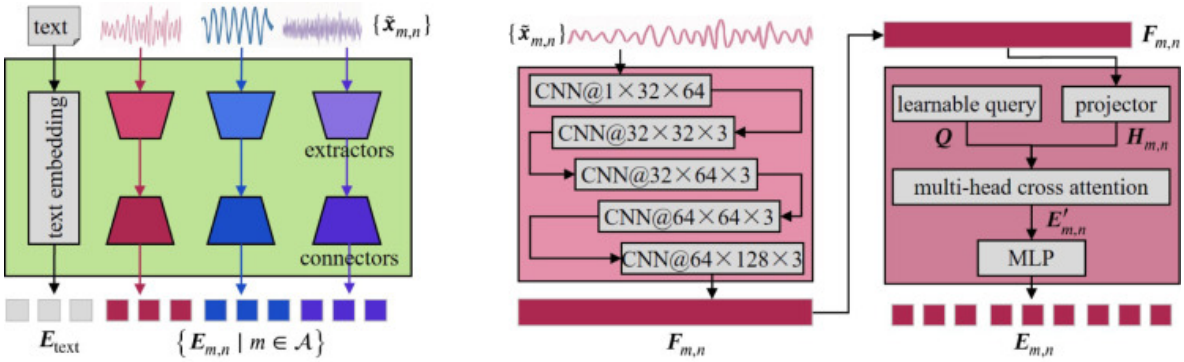
ntations that LLMs can understand and process. As

shown in Fig. 4, this paper adopts a multisensor embedding layer to handle multisensor signal embedding. The embedding layer consists of two main components: a feature extractor and a feature alignment module.



(a) Multisensor embedding layer (b) Structure of extractor and connector

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Fig. 4. Structure of the multisensor embedding layer.

3.2.1. Pre-processing and feature extractors

Before embedding multisensor signals, it is necessary to perform preprocessing on the raw signals. The goal of preprocessing is to convert different types of sensor signals into a unified format to facilitate subsequent feature extraction and alignment. Assume there are M types of sensors (e.g., vibration, acoustic, current, etc.), and the mth type includes N_m specific sensors. Here, $m = 1, 2, \ldots, M$, and $x_{m,n}$ denotes the raw time-domain signal from the nth sensor of the mth type, where $n = 1, 2, \ldots, N_m$. Since sensors within the same category may have different sampling rates and sensitivities, each signal is first normalized by resampling and power normalization to reduce the influence of varying operating conditions. The process is defined as:

$$x_{m,n}^{(r)} = \text{Resample}\left(x_{m,n}\right) \tag{3}$$

$$\tilde{x}_{m,n} = \frac{x_{m,n}^{(r)}}{\sqrt{\frac{1}{T_r}\sum_{t=1}^{T_r} x_{m,n}^{(r)}[t]^2}} \tag{4}$$

where $\text{Resample}(\cdot)$ denotes resampling to a unified sampling rate, and $\tilde{x}_{m,n}$ is the final normalized signal. T_r denotes the length of the signal after resampling.

Various methods exist for processing raw multisensor signals, including time-domain, frequency-domain, time-frequency analysis, and neural network-based techniques. In this work, each type of sensor is assigned a dedicated feature encoder network to extract features from its time-domain signals. The feature extraction process is given by:

$$F_{m,n} = \text{Ext}_m\left(\tilde{x}_{m,n}\right) \in R^{B \times T_{\text{in}} \times d_{\text{in}}} \tag{5}$$

where Ext_m is the feature extractor for the mth sensor type, and $F_{m,n}$ denotes the extracted feature representation. Here, B is the batch size, T_{in} is the input sequence length, and d_{in} is the input feature dimension.

Although a variety of encoder architectures can be used, including traditional signal processing techniques and deep neural networks, this paper mainly focuses on the LLM-based fusion and diagnosis. Therefore, the internal structure of the feature encoders is not the primary focus. In the implementation, a five-layer CNN operating on time-domain signals is utilized as the feature extractor, as illustrated on the left side of Fig. 4(b).

3.2.2. Features alignment connector

In multisensor data, different types of signals convey distinct physical meanings. For instance, vibration signals

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rrent signals mainly represent the load conditions of

machinery. Therefore, when performing feature alignment across sensor types, it is necessary to account for the inherent physical differences among signals. Moreover, since LLMs typically operate on textual input, the extracted features from different sensor types must be mapped into a semantic space that is interpretable by the LLM. As illustrated on the right side of Fig. 4(b), a connector network is used to align the multisensor features with the LLM's input space. This network transforms each feature representation into an embedding that matches the token space used by the LLM.

Specifically, for each feature tensor $F_{m,n}$ extracted from the nth sensor of the mth sensor type, a linear projection is first applied to match the hidden dimension d_{hid} of the LLM:

$$H_{m,n} = F_{m,n} \cdot W_{\text{proj}}^{\top} + b_{\text{proj}} \in R^{B \times T_{\text{in}} \times d_{\text{hid}}} \tag{6}$$

where W_{proj} and b_{proj} denote the weight and bias of the projection layer, respectively. Next, a learnable set of query vectors $q \in R^{N_q \times d_{\text{hid}}}$ is defined, and repeated across the batch dimension to obtain a query tensor:

$$Q = \text{Repeat}\left(q, B\right) \in R^{B \times N_q \times d_{\text{hid}}} \tag{7}$$

Using Q as the query and $H_{m,n}$ as both key and value, a multi-head attention module is applied to generate intermediate embeddings:

$$E'_{m,n} = \text{MultiHeadAttn}\left(Q, H_{m,n}, H_{m,n}\right) \in R^{B \times N_q \times d_{\text{hid}}} \tag{8}$$

where $E'_{m,n}$ represents the refined representation of the nth sensor in the mth sensor type, and N_q is the number of learnable queries. The attention mechanism is implemented following the standard multi-head attention described in [29].

Subsequently, a lightweight Multiple Layer Perceptron (MLP) is applied to further transform $E'_{m,n}$ into the final aligned embedding:

$$E_{m,n} = \text{MLP}\left(E'_{m,n}\right) = \sigma E'_{m,n} W_1 + b_1 W_2 + b_2 \tag{9}$$

where $W_1 \in R^{d_{\text{hid}} \times d_{\text{ff}}}$, $W_2 \in R^{d_{\text{ff}} \times d_{\text{hid}}}$, b_1, b_2 are bias parameters, d_{ff} is the feed-forward dimension, and $\sigma(\cdot)$ denotes a non-linear activation function. The MLP refines the attention outputs with non-linear transformations and provides stable embeddings for initializing subsequent expert sub-networks. Through supervised learning with paired signal-text diagnostic descriptions, the connector is trained to map heterogeneous sensor features into a unified semantic space, ensuring consistent alignment across signals and providing optimized initialization for downstream expert networks.

Let $A \subseteq \{1, 2, \ldots, M\}$ denote the set of sensor types present in the current input. The final multimodal embedding is formed by concatenating all available sensor features with the token embeddings of the textual prompt:

$$E_{\text{multi}} = \text{Concat}\, E_{\text{text}}, E_{m,n} \mid m \in A \tag{10}$$

where $\text{Concat}(\cdot)$ denotes sequence-wise concatenation. Multimodal embedding $E_{\text{multi}} \in R^{B \times N_t + \sum_{m \in A} N_{qm} \times d_{\text{hid}}}$ is then fed into the LLM for downstream fault diagnosis, where N_t is the number of text tokens and N_{qm} is the number of query vectors for sensor type m.

3.3. MoE-LLM

3.3.1. Expert sub-networks

Traditional LLMs are primarily designed for language processing tasks and lack the capability to deeply understand and extract features from sensor signals in fault diagnosis scenarios. To address this limitation, a sparse MoE mechanism is incorporated into the LLM, as illustrated in Fig. 5, enabling dynamic activation of expert sub-networks based on arbitrary combinations of sensor inputs. The MoE mechanism divides the LLM into multiple

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expert sub-networks and selects appropriate experts adaptively according to the input features, thereby achieving sparse activation and adaptive feature extraction.

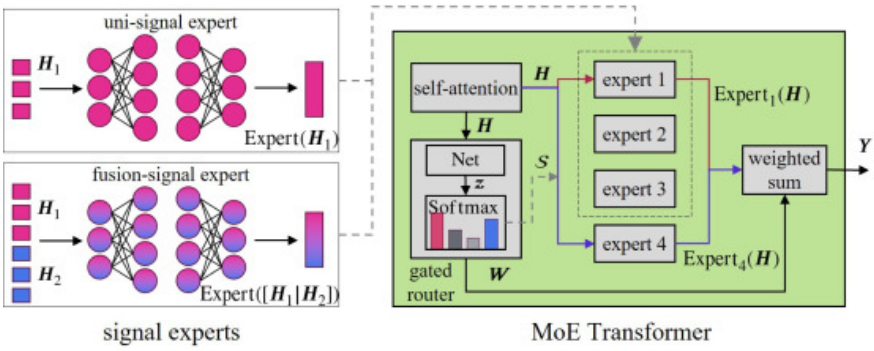

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Fig. 5. The structure of the MoE transformer.

To enable signal-aware specialization, a total of $2^M - 1$ expert sub-networks are constructed for M sensor types, with each expert corresponding to a specific combination of sensor signals. For example, in the case of M=2 (e.g., vibration and current), two uni-signal experts and one fusion-signal expert are defined: Expert 1 corresponds to vibration-only input, Expert 2 to current-only input, and Expert 3 to the combination of vibration and current. As illustrated on the left side of Fig. 5, the uni-signal expert processes features from a single type of signal, while the fusion-signal expert handles concatenated features derived from multiple signal types. To retain the original language modeling capability of the LLM, the standard feedforward network (FFN) in the Transformer block is preserved as a text expert (Expert 4), resulting in a total of 2^M expert sub-networks.

3.3.2. Sparse MoE mechanism

The sparse MoE mechanism replaces the standard FFN layer in selected Transformer blocks with a modular signal-adaptive structure, as shown in the right part of Fig. 5. The process is summarized as follows:

(1) Let H denote the output from the self-attention layer. This representation is passed into a router module, which computes a score vector z via a network $\mathrm{Net}(\cdot)$ and then applies a softmax to obtain expert weights $W = \left\{w_1, w_2, \ldots, w_{2^M}\right\}$:

$$z = \mathrm{Net}(H) \tag{11}$$

$$W = \mathrm{Softmax}(z) \tag{12}$$

where $w_i = e^{z_i} / \sum_{j=1}^{2^M} e^{z_j}$ represents the normalized gating weight for the ith expert, and $\mathrm{Net}(\cdot)$ is implemented as a simple linear transformation.

(2) The top-K experts with the highest weights are selected to form the active expert set S:

$$S = \mathrm{TopK}(W, K) \tag{13}$$

where $\mathrm{TopK}(\cdot, K)$ selects K elements with the highest weights from the gating vector W, and K is a hyperparameter controlling the number of active experts.

(3) Each selected expert Expert_i receives the same input H, and the final output is obtained by performing a weighted aggregation:

$$Y = \sum_{i \in S} w_i \cdot \mathrm{Expert}_i(H) \tag{14}$$

The sparse activation mechanism improves efficiency by activating only the top-K experts instead of all 2^M. This

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d_{hid}^2) with $K < 2^M$, lowering both computation and

memory cost. Meanwhile, the unused experts remain idle, allowing the model to maintain large capacity without proportional runtime overhead. To control computational cost and memory usage, the MoE structure is not applied to all Transformer layers, but only to selected ones throughout the model.

3.4. Training strategy

A curriculum learning-based training strategy is adopted, which progressively increases task complexity and enhances the model's generalization ability. As illustrated in Fig. 6 and Algorithm 1, the training process is divided into the following four stages:

Algorithm 1 Training Procedure of MoE-LLM

Input: Preprocessed samples $\{(\tilde{\boldsymbol{x}}, \boldsymbol{y}, \boldsymbol{y}_{\text{text}})_{m,n,i}\}$
Output: Trained MoE-LLM model

(1) Pre-processing and pre-train feature extractors
for each m **do**
  while not converged **do**
    Sample batch $\mathcal{B}_m \subset \{(\tilde{\boldsymbol{x}}, \boldsymbol{y})_{m,n,i}\}$
    Update $\boldsymbol{\theta}_m$ using Eq. (15) and Eq. (16)
  end while
end for
(2) Alignment features and fault descriptions
for each m **do**
  Freeze Ext_m and LLM
  while not converged **do**
    Sample batch $\mathcal{B}_m \subset \{(\tilde{\boldsymbol{x}}, \boldsymbol{y}_{\text{text}})_{m,n,i}\}$
    Update $\boldsymbol{\phi}_m$ using Eq. (17) and Eq. (18)
  end while
end for
(3) Fine-tune experts
for each $s \in \{1, \ldots, 2^M - 1\}$ **do**
  Freeze $\{\text{Ext}_m\}$, $\{\text{Conn}_m\}$, and LLM (except Expert_s)
  while not converged **do**
    Sample batch $\mathcal{B}_s \subset \{(\{\tilde{\boldsymbol{x}}_m\}_{m=1}^{|C_s|}, \boldsymbol{y}_{\text{text}})_{n,i}\}$
    Update $\boldsymbol{\psi}_s$ using Eq. (19) and Eq. (20)
  end while
end for
(4) Fine-tune MoE-LLM for specific task
Replace FFN layers in selected Transformer blocks with MoE layers
Freeze $\{\text{Ext}_m\}$ and $\{\text{Conn}_m\}$
Insert LoRA modules with $\boldsymbol{A}, \boldsymbol{B}$
while not converged **do**
  Sample batch $\mathcal{B}_{\text{task}} \subset \{(\tilde{\boldsymbol{x}}, \boldsymbol{y}_{\text{text}})_{m,n,i}\}$
  Update $\boldsymbol{A}, \boldsymbol{B}$ using Eq. (21) and Eq. (22)
end while

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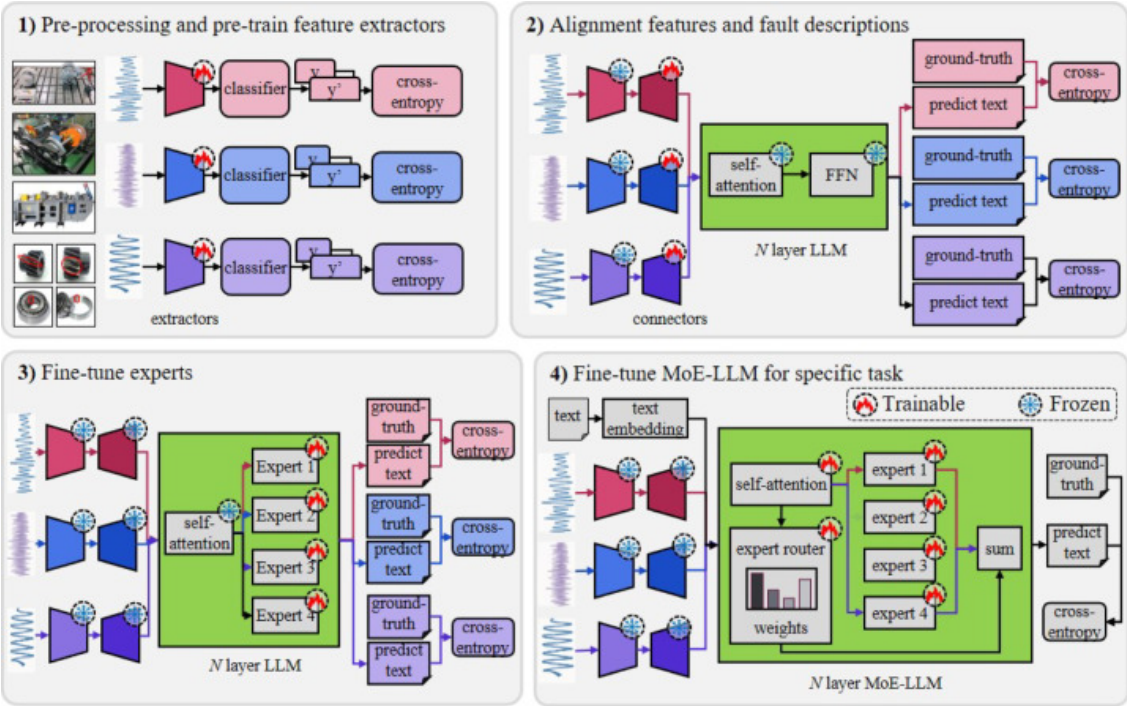

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Fig. 6. The training strategy of the proposed method.

(1) Pre-processing and pre-train feature extractors. Raw time- domain signals from different sensor types are preprocessed, including sampling rate normalization and power normalization. For each sensor type m, a pretraining sub-network is constructed, composed of a feature extractor Ext_m and a simple classifier Clf_m. Each training batch $B \subset \tilde{x}_{m,n,i} \mid n = 1, \ldots, N_m, \quad i = 1, \ldots, N_s$ is used as input, where N_m is the number of sensors of type m, and N_s is the number of signal samples per sensor. The feature extractor Ext_m is optimized by minimizing the cross-entropy loss between the classifier output y' and the ground-truth label y:

$$L_{\mathrm{CE}} = -\frac{1}{N_b}\sum_{j=1}^{N_b}\sum_{c=1}^{C} y_{j,c} \log y'_{j,c} \tag{15}$$

$$\theta_m \leftarrow \theta_m - \eta \cdot \frac{\partial L_{\mathrm{CE}}}{\partial \theta_m} \tag{16}$$

where L_{CE} is the cross-entropy loss, N_b is the batch size, C is the number of fault classes, and θ_m denotes the learnable parameters of Ext_m, and η is the learning rate.

(2) Alignment features and fault descriptions. After the feature extractor is pretrained, its parameters are frozen. The output of Ext_m is passed to the connector network Conn_m, which generates an embedding compatible with the LLM. The embedding is then input to a frozen LLM to generate the corresponding fault description. The connector Conn_m is optimized by minimizing the token-level cross-entropy loss between the predicted sequence $\hat{y}_{\mathrm{text}} = \left(\hat{y}_1, \ldots, \hat{y}_T\right)$ and the ground-truth sequence $y_{\mathrm{text}} = \left(y_1, \ldots, y_T\right)$, where T is the length of the text sequence:

$$L_{\mathrm{text}} = -\frac{1}{T}\sum_{t=1}^{T}\sum_{k=1}^{V} y_{t,k} \log \hat{y}_{t,k} \tag{17}$$

$$\phi_m \leftarrow \phi_m - \eta \cdot \frac{\partial L_{\mathrm{text}}}{\partial \phi_m} \tag{18}$$

where ϕ_m denotes the parameters of the connector network Conn_m, V is the vocabulary size of LLM (i.e., the total number of unique tokens the language model can generate), and $y_{t,k}$ is a one-hot encoding of the kth token at position t.

(3) Fine-tune experts. In this stage, the pretrained expert modules are used to replace the FFN modules in selected Transformer layers. To ensure training stability, parameters of Ext_m, Conn_m, and all components of the LLM

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ing iterations, every possible non-empty sensor combination C_s is selected in turn. Meanwhile, the corresponding expert Expert_s is activated. The optimization

process is:

$$L_{\text{expert}} = -\frac{1}{T}\sum_{t=1}^{T}\sum_{k=1}^{V} y_{t,k} \log \hat{y}_{t,k}^{(s)} \tag{19}$$

$$\psi_s \leftarrow \psi_s - \eta \cdot \frac{\partial L_{\text{expert}}}{\partial \psi_s} \tag{20}$$

where $\hat{y}_{t,k}^{(s)}$ is the predicted token probability contributed by the expert Expert_s, and ψ_s denotes the parameters of the expert sub-network corresponding to the sensor combination C_s. The loss function L_{expert} is computed based on the output of the activated expert, and the parameters are updated accordingly.

(4) Fine-tune MoE-LLM for specific task.

After expert tuning, the MoE layers are formally integrated into selected Transformer blocks, replacing the original FFN modules to enhance multisensor understanding. During this stage, the parameters of Ext_m, Conn_m are kept frozen. Only attention modules, expert routing modules, and expert sub-networks, are fine-tuned. To mitigate the heavy computational cost of full-parameter tuning, Low-Rank Adaptation (LoRA) [36], a widely adopted lightweight fine-tuning technique, is used. The key idea of LoRA is to approximate the update of large weight matrices by a low-rank decomposition, which greatly reduces the number of trainable parameters. This not only lowers memory consumption but also mitigates the risk of overfitting, making it especially suitable for task-specific adaptation of large LLMs.

Specifically, LoRA approximates the parameter update by a low-rank decomposition $W_{\text{LoRA}} = AB$, where $A \in R^{d \times r}$ and $B \in R^{r \times d}$, with $r \ll d$. The training is conducted on task-specific datasets containing samples that cover all $2^M - 1$ possible non-empty combinations of sensor. During training, the original weight W_0 remains frozen, and the final effective weight becomes $W = W_0 + W_{\text{LoRA}}$ as shown in Fig. 7. This strategy enables efficient adaptation while significantly reducing memory consumption and overfitting risk. During backpropagation, only the low-rank matrices A and B are updated using gradient descent:

$$L_{\text{LLM}} = -\frac{1}{T}\sum_{t=1}^{T}\sum_{k=1}^{V} y_{t,k} \log \hat{y}_{t,k}^{(\text{LoRA})} \tag{21}$$

$$A \leftarrow A - \eta \cdot \frac{\partial L_{\text{LLM}}}{\partial A}, \quad B \leftarrow B - \eta \cdot \frac{\partial L_{\text{LLM}}}{\partial B} \tag{22}$$

where L_{LLM} is the loss function for the LLM fine-tuning, and $\hat{y}_{t,k}^{(\text{LoRA})}$ is the predicted token probability from the LoRA-modified LLM.

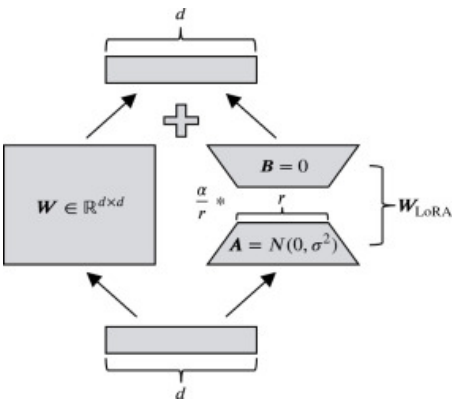

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Fig. 7. The LoRA mechanism for efficient fine-tuning of LLMs.

Through this four-stage training process, the MoE-LLM is progressively optimized for cross-signal representation learning and diagnostic text generation.

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4.1. Datasets

Three multisensor rotating machinery datasets are utilized in this study: the Intelligent Bearing (IB) dataset [37], the BJTU-A dataset (a subset of the BJTU train bogie dataset) [38], and the Motor Bearing (MB) dataset [39], as illustrated in Fig. 8(a), Fig. 8(b), and Fig. 8(c), respectively. These datasets include various types of sensor signals such as vibration, acoustic, and current signals, with sampling rates ranging from 12 kHz to 64 kHz. They also contain a variety of health conditions and operating scenarios.

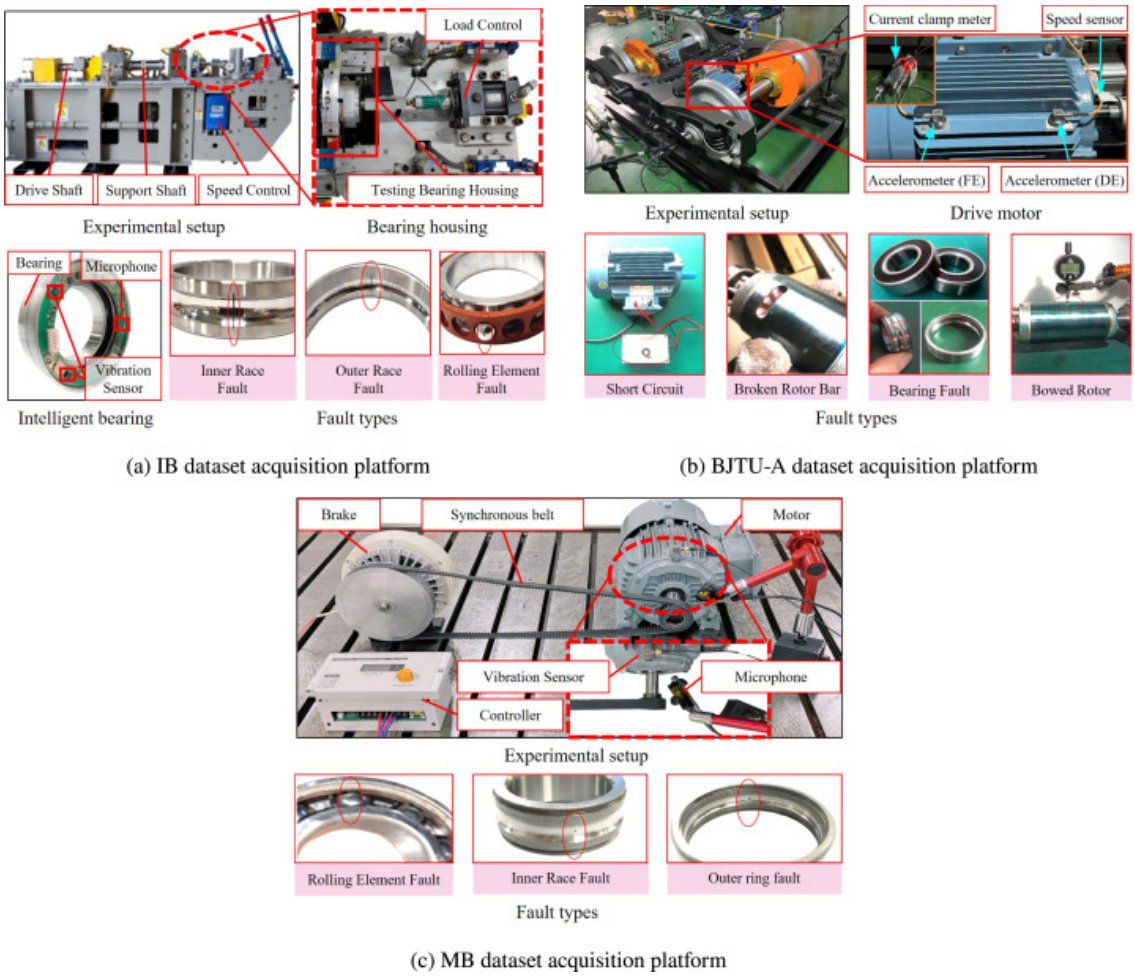

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Fig. 8. Test rigs used for data acquisition in each dataset.

The IB dataset includes vibration signals from both vertical and horizontal directions, as well as acoustic signals, collected under three rotational speeds and three load conditions. The vibration signals are sampled at 32 kHz, and the acoustic signals at 48 kHz. The dataset covers various bearing fault types, including normal, rolling element faults, inner race faults, and outer race faults, each with three severity levels: minor, moderate, and severe. Labels range from 0 to 9. Each sample contains 0.1 s of vibration and acoustic data.

In the BJTU dataset, the subset relevant to the drive motor is extracted, referred to as the BJTU-A dataset. It includes six channels of triaxial vibration signals from the drive end and fan end, and three-phase current signals collected via current clamp meters. All signals are sampled at 64 kHz. The dataset includes five fault conditions: normal, short circuit, broken rotor bar, bearing fault, and bowed rotor, with labels ranging from 0 to 4. Each sample contains 0.05 s of vibration and current signals.

The MB dataset contains six channels of triaxial vibration signals from the drive end and fan end of a motor bearing, along with one channel of acoustic signal collected by a microphone. All signals are sampled at 12 kHz. Fault types include normal, rolling element faults, inner race faults, and outer race faults. Except for the inner race

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fault, which lacks a moderate severity class, all others are divided into three severity levels. Labels range from 0 to 8. Each sample contains 0.1 s of vibration and acoustic data.

The class distributions of the three datasets are presented in Table 1.

Table 1. Class distribution of the three datasets.

Dataset	Label	Train num	Test num
IB	0	1575	173
	1	1773	183
	2	1749	191
	3	1560	182
	4	1485	171
	5	1395	138
	6	1365	140
	7	1593	197
	8	1353	139
	9	1380	178
BJTU-A	0	3240	360
	1	3240	360
	2	3240	360
	3	3240	360
	4	3240	360
MB	0	4860	540
	1	4860	540
	2	4752	528
	3	4320	480
	4	4860	540
	5	4860	540
	6	4428	492
	7	4860	540
	8	4860	540

4.2. Implementation details

For each dataset, the constructed samples, prompts, and labels are illustrated in Fig. 9. Each sample prompt includes a description of the sensor signals and the corresponding fault description. The label corresponds to the fault diagnosis or treatment response.

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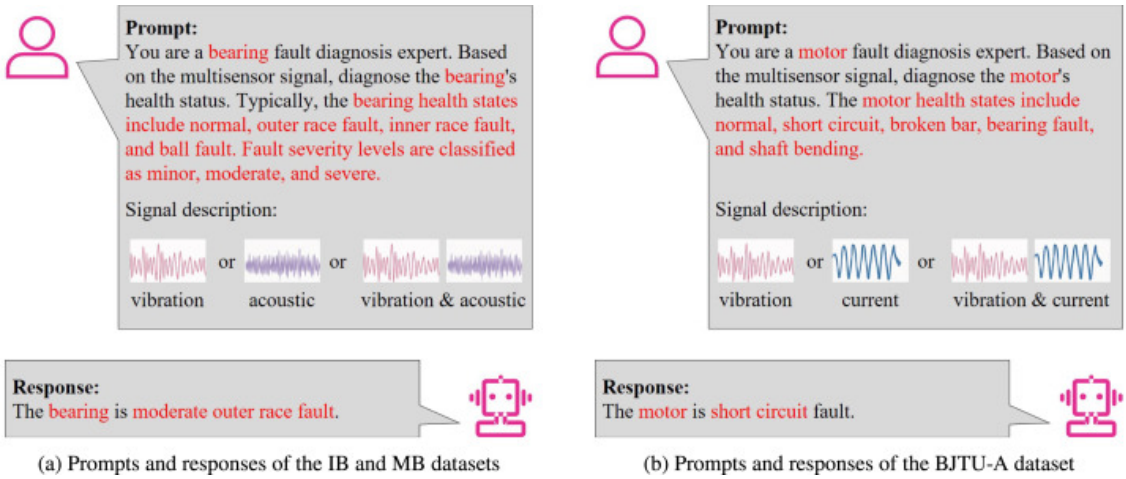

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Fig. 9. Examples of prompts and responses in each dataset.

The base LLM employed in this study is DeepSeek-R1-Distill-Qwen-1.5B [40], which contains 1.5 billion parameters and consists of 27 Transformer layers. Considering both performance and efficiency, five Transformer layers (layers 4, 8, 12, 16, and 20; zero-indexed) are modified with MoE structures. In the multisensor embedding layer, the feature extractor is implemented using a 5-layer CNN for time-domain feature extraction, with an output dimension of 128. The connector network is implemented as a simple linear projection that maps features from 128 to the LLM's token embedding dimension of 1536. Each expert sub-network within the MoE-LLM consists of one MLP module with a hidden size of 3072, as shown in Table 2, which shares the same structure as the final MLP layer in the connector. Each MoE-LLM Transformer layer includes four expert sub-networks, and K=2 experts are activated for each input. The low-rank dimension r of the LoRA module is set to 4. Model training is performed using the AdamW optimizer with an initial learning rate of 1×10^{-4} and a batch size of 32. All training and inference processes are conducted on a computing platform equipped with an Intel(R) Xeon(R) Gold 6248R CPU and NVIDIA GeForce RTX 4090 GPU, using the PyTorch and HuggingFace Transformers frameworks.

Table 2. Configuration of the MLP module.

Module	Input dimension	Output dimension	Bias
fully-connected (FC)	1536	3072	True
sigmoid linear unit (SiLU)	–	–	–
fully-connected (FC)	3072	1536	True

4.3. Diagnostic performance analysis

4.3.1. Comparative analysis with other methods

This section first introduces several baseline multisensor fusion strategies. Specifically, Fusion-Add aggregates multisensor features by element-wise summation, Fusion-Con concatenates feature vectors across signals, and Fusion-Attn applies a lightweight self-attention mechanism for feature interaction. All these approaches use the same feature extractors as described previously. In addition, two recently proposed multisensor fusion methods, MCFormer [18] and res-GTLA-Net [19], are considered.

For further comparison, several existing Transformer-based fault diagnosis approaches, including SiT [41], TST [42], DFormer [43], TAR [44], and LLM-based methods such as LLMFB-1 [27], LLMFB-2 [27], and BearLLM [45] are also

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descriptions of time-domain and frequency-domain

features, LLMFB-2 tokenizes the signals and feeds them as embedded sequences with positional encoding, and BearLLM applies a CNN-based extractor on DCT-transformed signals. For fair comparison, BearLLM's original feature extractor is replaced with the proposed one. All methods are trained and tested under identical experimental conditions.

The diagnostic results are shown in Table 3. The proposed method consistently achieves the best or near-best accuracy and F1-score across all three datasets, with scores of 97.87%/97.85%, 99.76%/99.76%, and 99.95%/99.95%, respectively. Compared with multisensor fusion approaches, the proposed method generally outperforms them, while maintaining greater flexibility. In contrast, these baselines rely on fixed fusion forms and rigid input combinations, limiting their adaptability to diverse sensor configurations.

Table 3. Diagnosis performance comparison.

Method	IB dataset		BJTU-A dataset		MB dataset	
	Acc (%)	F1 (%)	Acc (%)	F1 (%)	Acc (%)	F1 (%)
Fusion-Add	94.56 ± 0.68	94.93 ± 0.69	97.93 ± 0.22	97.92 ± 0.23	97.99 ± 0.03	97.97 ± 0.03
Fusion-Con	95.27 ± 0.71	95.57 ± 0.68	97.46 ± 0.07	97.46 ± 0.07	95.27 ± 0.08	95.21 ± 0.08
Fusion-Attn	97.30 ± 0.48	97.50 ± 0.45	96.98 ± 0.52	96.98 ± 0.52	94.23 ± 0.67	94.23 ± 0.67
MCFormer [18]	90.46 ± 0.34	91.24 ± 0.33	97.69 ± 0.37	97.68 ± 0.37	97.41 ± 0.36	97.42 ± 0.35
Res-GTLA-Net [19]	93.75 ± 1.10	94.16 ± 1.13	99.44 ± 0.08	99.44 ± 0.08	99.18 ± 0.27	99.18 ± 0.27
SiT [41]	79.08 ± 1.19	79.00 ± 1.20	96.11 ± 0.09	96.11 ± 0.09	97.60 ± 0.12	97.60 ± 0.12
TST [42]	90.01 ± 0.84	90.45 ± 0.82	89.83 ± 1.03	89.81 ± 1.01	93.98 ± 0.31	94.01 ± 0.31
DFormer [43]	32.88 ± 0.63	32.90 ± 0.44	96.28 ± 1.19	96.28 ± 1.18	95.14 ± 0.10	95.10 ± 0.10
TAR [44]	97.38 ± 0.27	97.58 ± 0.27	99.96 ± 0.03	99.96 ± 0.03	99.89 ± 0.03	99.89 ± 0.03
LLMFB-1 [27]	73.25 ± 0.15	73.23 ± 0.14	95.33 ± 0.18	95.34 ± 0.18	49.05 ± 0.34	45.52 ± 0.36
LLMFB-2 [27]	79.64 ± 1.05	80.19 ± 1.22	97.02 ± 0.16	97.01 ± 0.16	98.86 ± 0.03	98.87 ± 0.03
BearLLM [45]	82.29 ± 0.10	82.01 ± 0.15	97.43 ± 0.16	97.42 ± 0.16	95.02 ± 0.05	94.99 ± 0.05
Proposed	97.87 ± 0.22	97.85 ± 0.22	99.76 ± 0.03	99.76 ± 0.03	99.95 ± 0.01	99.95 ± 0.01

Compared with Transformer-based methods and existing LLM-based approaches, on the IB dataset, traditional Transformer models such as SiT and TST yield F1-scores of 79.00% and 90.45%, respectively, while DFormer performs poorly with only 32.90%. In contrast, the proposed method achieves a much higher F1-score of 97.85%, improving by more than 18 percentage points. On the BJTU-A dataset, TST achieves 89.81%, and LLMFB-2 reaches 97.01%, while the proposed approach further improves to 99.76%, showing its superior fine-grained recognition ability in industrial motor diagnostics. On the MB dataset, although LLMFB-2 and TAR perform strongly with F1-scores of 98.87% and 99.89%, respectively, the proposed method still surpasses them slightly with 99.95%, demonstrating robust adaptability to multisensor fusion inputs.

4.3.2. Analysis of classification performance

The confusion matrices for the proposed method on different datasets are presented in Fig. 10. As observed, the method achieves accurate classification across fault categories with low misclassification rates, further confirming its high diagnostic accuracy and strong generalization capability. On the IB dataset, the few misclassifications mainly occur between different severities of rolling element faults and the normal state, which can be attributed to the relatively weak fault signatures of ball defects. On the BJTU-A dataset, the primary confusion arises from

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On the MB dataset, the misclassification patterns include

confusion among different severities of outer race faults and occasional misidentification of inner race faults as ball faults, the latter likely due to the absence of severe inner race fault samples.

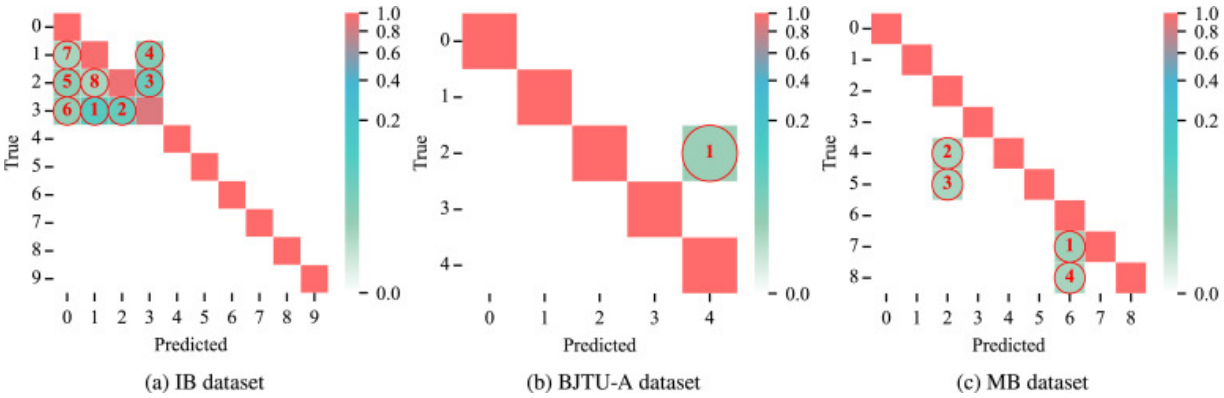

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Fig. 10. Confusion matrices of the proposed method on different datasets (The red circles with numbers indicate the misclassification cases at the corresponding positions, where the numbers represent their ranking from the largest to the smallest according to the number of occurrences.).

4.3.3. Performance under different input signals

Another notable advantage of the proposed method lies in its flexibility to handle various combinations of sensor inputs. The diagnostic accuracy under different sensor combinations is presented in Table 4. It can be observed that the proposed model achieves consistently high performance across all signal configurations.

Table 4. Accuracy of the proposed method under different input signals (V: Vibration, A: Acoustic, C: Current).

Signal	IB dataset		BJTU-A dataset		MB dataset	
	Acc (%)	F1 (%)	Acc (%)	F1 (%)	Acc (%)	F1 (%)
V	95.78 ± 0.23	95.73 ± 0.23	99.37 ± 0.09	99.37 ± 0.09	95.68 ± 0.12	95.74 ± 0.12
A	83.08 ± 0.42	83.08 ± 0.46	–	–	99.70 ± 0.04	99.70 ± 0.04
C	–	–	76.43 ± 0.49	75.58 ± 0.46	–	–
V+A	97.87 ± 0.22	97.85 ± 0.22	–	–	99.95 ± 0.01	99.95 ± 0.01
V+C	–	–	99.76 ± 0.03	99.76 ± 0.03	–	–

On the IB dataset, the combination of vibration and acoustic signals yields an accuracy of 97.87%, which is significantly higher than using vibration or acoustic signals alone. For the BJTU-A dataset, the combination of vibration and current signals achieves 99.76% accuracy, clearly outperforming single-signal inputs. Similarly, for the MB dataset, combining vibration and acoustic signals results in an accuracy of 99.95%, again demonstrating notable improvement over individual signals.

In summary, the proposed MoE-LLM model can dynamically activate appropriate expert sub-networks based on the input sensor configuration, thereby effectively leveraging the complementary characteristics of multisensor information. This leads to significantly enhanced diagnostic accuracy and robustness across various application scenarios.

4.3.4. Performance under domain shift

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domain shifts are more common in real-world applications. When a model trained using sensor data from one measurement location is applied to another

location, a significant performance drop may occur due to domain shift, i.e., the difference in data distributions across locations [46]. A common solution is transfer learning via fine-tuning, where the model is trained on data from one domain and then adapted using a small subset of samples from the target domain.

In this study, domain shift experiments are conducted on the IB, BJTU-A, and MB datasets using vibration signals, which are the most representative sensor type for fault diagnosis. Other signal types are kept unchanged to provide stable complementary information. This setting simulates the practical case where a vibration sensor is mounted at a different location (e.g., vertical vs. horizontal axis or drive end vs. fan end), resulting in domain shift on the vibration channel while other sensors remain unaffected. Specifically, for the IB dataset, the model is trained on vertical-axis vibration signals and fine-tuned with only 10% of horizontal-axis data. For the BJTU-A dataset, vibration data from the drive end are used for training, followed by fine-tuning using 10% of fan-end data. Similarly, for the MB dataset, training is conducted on drive-end vibration signals and fine-tuned using 10% of fan-end samples.

The diagnostic performance of several LLM-based methods under domain shift is reported in Table 5. The proposed method achieves high accuracy/F1-score across all datasets, with 96.87%/96.85% on IB, 97.78%/97.78% on BJTU-A, and 99.97%/99.97% on MB for accuracy and F1-score, respectively. Compared to other methods, the proposed approach consistently outperforms alternatives, demonstrating its robustness in handling domain shifts. Notably, the model requires only a small number of fine-tuning samples to adapt effectively to new sensor distributions.

Table 5. Diagnosis accuracy of experiments with domain shift .

Method	**IB dataset**		**BJTU-A dataset**		**MB dataset**	
	Acc (%)	**F1 (%)**	**Acc (%)**	**F1 (%)**	**Acc (%)**	**F1 (%)**
LLMFB-1	50.73 ± 0.71	50.35 ± 0.75	54.68 ± 0.29	54.51 ± 0.12	34.67 ± 0.17	32.57 ± 0.14
LLMFB-2	56.09 ± 0.71	54.74 ± 1.10	81.92 ± 0.03	81.70 ± 0.06	82.56 ± 0.54	82.82 ± 0.43
BearLLM	36.51 ± 0.75	36.10 ± 0.91	51.46 ± 0.52	51.52 ± 0.44	36.93 ± 0.30	34.97 ± 0.36
Proposed	96.87 ± 0.17	96.85 ± 0.17	97.78 ± 0.02	97.78 ± 0.02	99.97 ± 0.01	99.97 ± 0.01

4.4. Ablation study

4.4.1. Effectiveness of MoE layers

An ablation study is conducted on the IB dataset to evaluate the effectiveness of integrating MoE layers. As shown in Table 6, the model with MoE layers achieves higher accuracy and F1-score across all input signal configurations. Notably, for vibration signals (V), accuracy increases by 1.26% and F1-score improves by 1.34%, indicating that the MoE mechanism significantly enhances diagnostic performance.

Table 6. Ablation study on the effectiveness of MoE layers .

	Acc (%)	**F1 (%)**	**Acc (%)**	**F1 (%)**
Signal	**W/o MoE**		**With MoE**	
V	94.52 ± 0.49	94.39 ± 0.49	95.78 ± 0.23	95.73 ± 0.23
A	83.02 ± 0.22	83.07 ± 0.19	83.08 ± 0.42	83.08 ± 0.46
V+A	97.14 ± 0.24	97.10 ± 0.24	97.87 ± 0.22	97.85 ± 0.22
Avg	91.56	91.52	92.24	92.22

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Fig. 11 presents the training loss curves of the proposed method on the IB dataset. It can be observed that the model incorporating pretrained MoE layers converges faster and more stably, demonstrating its improved ability to adapt to the feature distributions of multisensor signals and enhanced training efficiency.

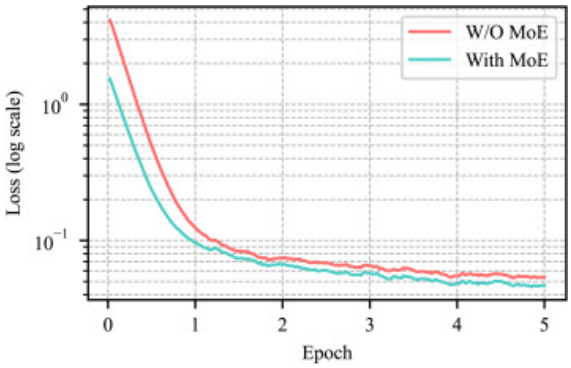

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Fig. 11. Training loss curve of the proposed method on the IB dataset.

4.4.2. Effectiveness of connector networks

The t-SNE visualization of vibration and acoustic features before and after alignment using the connector network is shown in Fig. 12. This experiment further verifies the effectiveness of the connector network. Before alignment, feature distributions of different categories are scattered with significant overlap in the feature space. After alignment, the features become more compact and well-separated, indicating that the connector network effectively optimizes feature distribution and enhances inter-class discrimination.

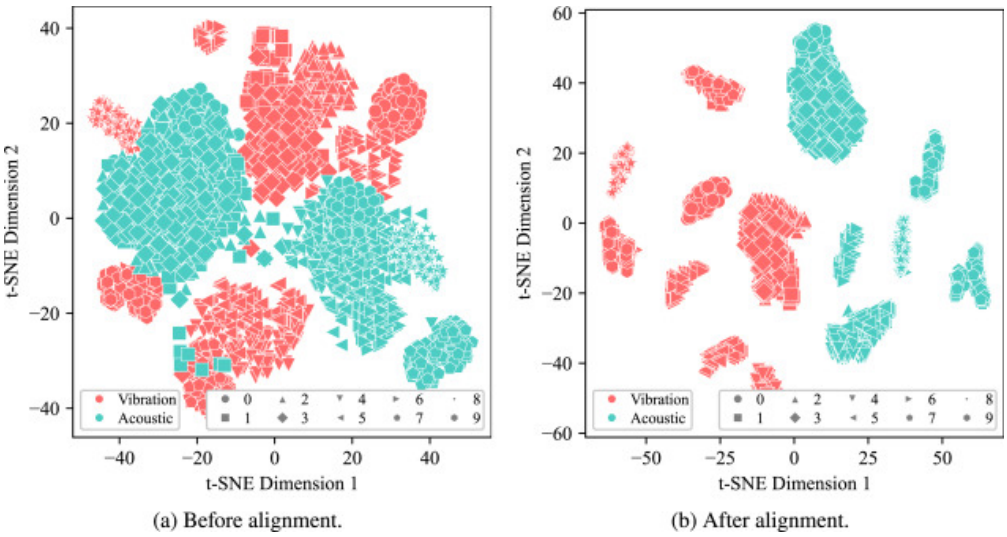

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Fig. 12. t-SNE visualization of the feature space before and after alignment.

4.4.3. Effectiveness of training strategies

To validate the effectiveness of the proposed training strategy, a series of ablation experiments were conducted. (1) End-to-end: The entire model, including the feature extractor and connector, was trained as a unified end-to-end system. (2) Proposed (w/o stage 3): The parameters of the expert networks were directly initialized using the final MLP parameters of the corresponding connector networks, while omitting the third-stage fine-tuning step. (3) Proposed (full staged training): the complete strategy. All methods were trained for one epoch.

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The experimental results are presented in Table 7. End-to-end training yielded lower accuracy since the parameters of the extractor, connector, and expert networks were not pre-trained, making it difficult for the model to converge. Compared with directly omitting stage 3 and proceeding to task-specific fine-tuning, the proposed method improved the average accuracy and F1-score across the three signal inputs from 82.93% and 82.53% to 89.24% and 88.95%, respectively. This demonstrates that the proposed training strategy can effectively enhance the model's performance.

Table 7. Accuracy under different training configurations on IB dataset .

Signal	End to end		Proposed (w/o stage 3)		Proposed	
	Acc (%)	F1 (%)	Acc (%)	F1 (%)	Acc (%)	F1 (%)
V	39.19 ± 1.13	38.57 ± 1.14	81.70 ± 0.61	81.13 ± 0.74	92.40 ± 0.36	92.31 ± 0.38
A	30.24 ± 0.94	30.24 ± 0.84	79.81 ± 0.90	79.45 ± 1.00	80.14 ± 0.05	79.49 ± 0.06
V+A	53.76 ± 0.49	53.74 ± 0.45	87.27 ± 0.39	87.00 ± 0.42	95.17 ± 0.25	95.05 ± 0.24
Avg	41.06	40.85	82.93	82.53	89.24	88.95

Furthermore, the incorporation of curriculum learning ensures that the model gradually adapts to increasing task complexity rather than directly tackling the final objective. By progressively training from general feature extraction to sensor type alignment and finally task-specific fine-tuning, the model not only achieves better feature representation but also exhibits stronger transferability to diverse fault scenarios. This staged learning paradigm enables the proposed method to be readily extended to other industrial applications where heterogeneous signals and task variations are present.

4.4.4. Effectiveness of text prompts

To verify the effectiveness of text prompts and investigate the impact of different prompts on model performance, three types of prompts, as illustrated in Fig. 13, were used to convert the input into tokens for the LLM tokenizer and evaluated on the IB dataset. Prompt 1 does not include any textual description, retaining only the start/end tokens and signal placeholders. Prompt 2 specifies the fault diagnosis task without restricting the output fault categories. Prompt 3 describes the diagnostic task while explicitly constraining the model to output within a predefined range of fault categories. All experiments were trained for one epoch.

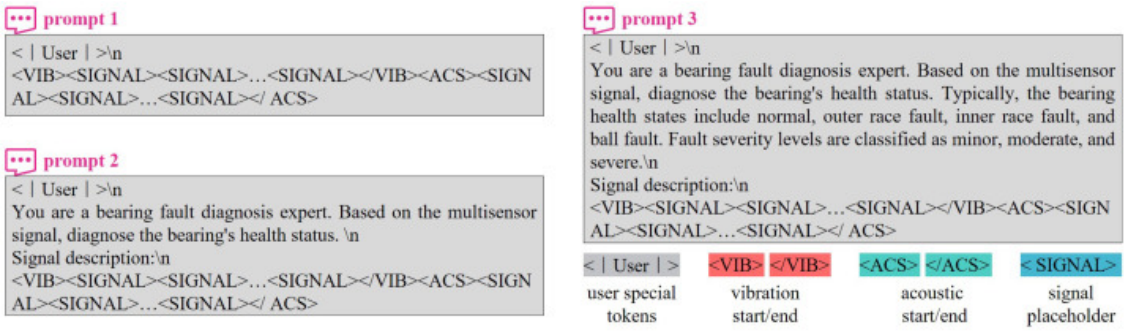

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Fig. 13. Different text prompts on IB dataset.

The results are reported in Table 8. The models trained with the three prompts achieved average accuracy and F1 of 87.74%/87.59%, 89.80%/89.59%, and 89.24%/88.95%, respectively. It can be observed that models with textual prompts significantly outperform those without prompts. Interestingly, the prompt without specifying fault signal and fused-signal inputs. This may be attributed to

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the relatively small scale of the adopted LLM, which imposes excessive cognitive burden when processing complex prompts. Specifically, the model must simultaneously handle both “fault type classification” and “severity classification”, leading to unstable training. In summary, incorporating text prompts leads to better performance on the IB dataset.

Table 8. Accuracy of the proposed method with different text prompts on IB dataset .

Signal	Prompt 1		Prompt 2		Prompt 3	
	Acc (%)	F1 (%)	Acc (%)	F1 (%)	Acc (%)	F1 (%)
V	88.20 ± 0.70	88.36 ± 0.70	91.61 ± 0.25	91.27 ± 0.26	92.40 ± 0.36	92.31 ± 0.38
A	80.59 ± 0.46	80.03 ± 0.58	82.07 ± 0.29	81.84 ± 0.31	80.14 ± 0.05	79.49 ± 0.06
V+A	94.44 ± 0.60	94.39 ± 0.61	95.71 ± 0.24	95.66 ± 0.25	95.17 ± 0.25	95.05 ± 0.24
Avg	87.74	87.59	89.80	89.59	89.24	88.95

4.5. Analysis of experts distribution

The distribution of activated experts under different signal combinations across datasets is illustrated in Fig. 14. On the IB and MB datasets, Expert 1 and Expert 2 correspond to vibration and acoustic experts, respectively. Expert 3 is responsible for fusing vibration and acoustic signals, while Expert 4 originates from the standard FFN layer and can be regarded as a general-purpose expert. On the BJTU-A dataset, Expert 1 and Expert 2 correspond to vibration and current experts, respectively, Expert 3 handles the fusion of vibration and current signals, and Expert 4 serves as the general transformer expert.

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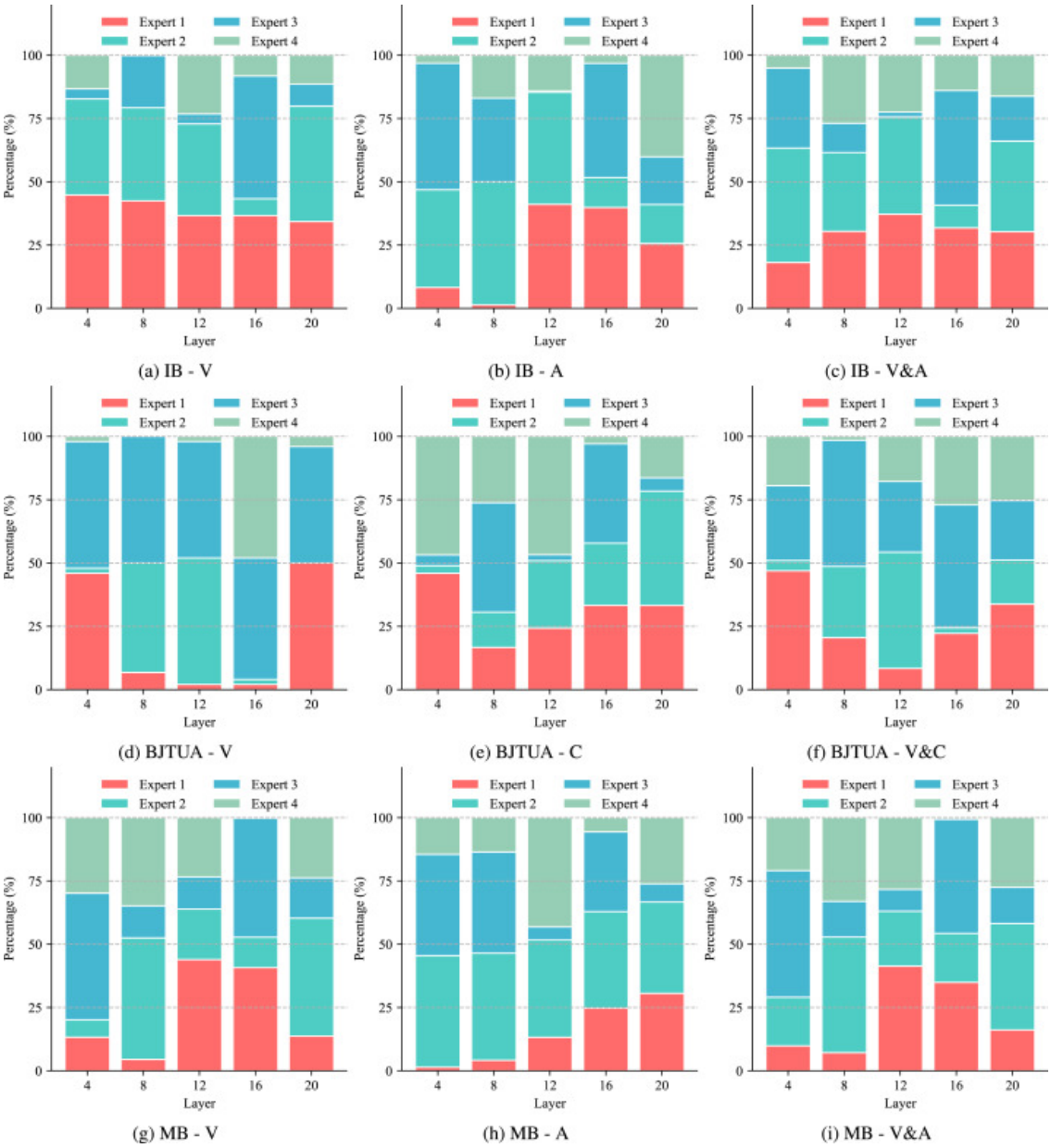

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Fig. 14. The distribution of experts in different signals.

On the IB dataset, when the input consists solely of vibration signals, Expert 1 exhibits the highest activation rate, indicating its specialization in processing vibration signals. When only acoustic signals are provided, Expert 2 becomes more active, especially at Layers 4 and 8, suggesting its role as the acoustic signal expert. When both vibration and acoustic signals are input, all experts are activated to varying degrees, showing that the model can jointly utilize information from both signal types for diagnosis.

On the BJTU-A dataset, when vibration signals are used as input, both Expert 1 and Expert 3 show high activation, indicating their effectiveness in handling vibration signals. When the input is current signals, Expert 2 is predominantly activated in deeper layers, while Expert 1 and Expert 4 are more active in shallower layers. This may be attributed to the fact that current signals are inherently more challenging for feature extraction, thus requiring deeper processing with support from the general-purpose Expert 4. When both vibration and current signals are provided, all experts are activated, implying effective multisensor feature fusion.

On the MB dataset, as the input transitions from vibration to acoustic signals, the activation of Expert 1 decreases while that of Expert 2 increases significantly, indicating the specialization of Expert 2 in processing acoustic signals. [illegible] nal and fusion-signal inputs, suggesting its capability to

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handle fused information. When both vibration and acoustic signals are input, a balanced activation across all experts is observed, demonstrating robust multisensor fusion capability.

To further analyze the selective activation property of expert networks and to investigate whether fault recognition errors are related to expert selection, we define the gating entropy to quantify the certainty of expert selection at each layer. The gating entropy is calculated using expert weights (as described in Eqs. (11)–(12)) as follows:

$$H = -\sum_{i=1}^{2^M} w_i \log\left(w_i\right) \tag{23}$$

A lower gating entropy indicates that the model selects experts with higher certainty, whereas a higher gating entropy suggests greater uncertainty in expert selection.

On the IB dataset, the distributions of gating entropy for correct and incorrect samples under different signal inputs are shown in Fig. 15. It can be observed that the gating entropy distributions of correct and incorrect samples do not exhibit significant differences, indicating that diagnostic errors are not directly associated with expert selection. Instead, the model automatically selects experts based on the type of input signals, while the correctness of diagnosis is determined by other factors. Furthermore, it can be observed that for all three input cases, the gating entropy at the 8th and 20th layers is generally lower, suggesting that expert selection is more deterministic at these layers. This may be attributed to the fact that these layers play a crucial role in feature extraction and fusion.

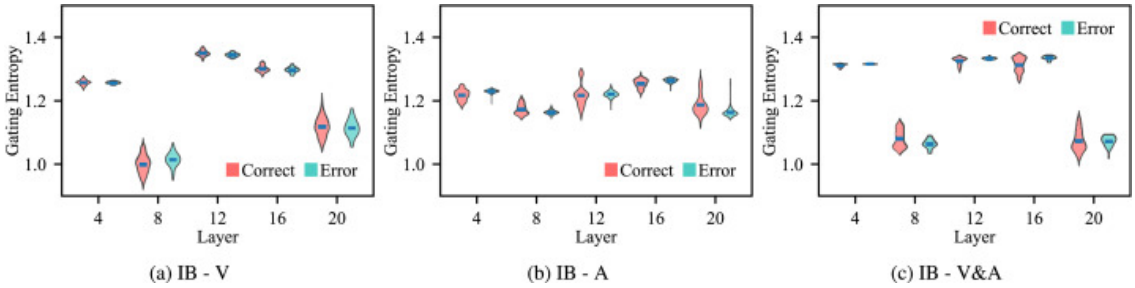

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Fig. 15. Gate entropy of different layers on the IB dataset.

In summary, the proposed method dynamically selects appropriate experts based on input signal types, which enhances diagnostic performance. The distribution patterns of expert activation under various signal combinations confirm the model's strong adaptability to multisensor data.

4.6. Hyper-parameters study

4.6.1. Impact of the value of K

Using the IB dataset as a case study, this section investigates the influence of the number of activated experts K in the MoE-LLM on diagnostic performance. As shown in Table 9, increasing K generally improves the model's accuracy and F1-score, though the gains diminish as K increases. The optimal performance is achieved when K=2, where the accuracy and F1-score reach 97.87% and 97.85%, respectively. Further increasing K to 3 slightly reduces performance, suggesting that while moderate expert activation improves representation learning, excessive expert activation may lead to redundancy and increased computational cost. Therefore, careful tuning of K is essential for balancing performance and efficiency.

Table 9. Impact of the value of K on the IB dataset .

Signal	$K=1$		$K=2$		$K=3$	
	Acc (%)	**F1 (%)**	**Acc (%)**	**F1 (%)**	**Acc (%)**	**F1 (%)**
				95.73 ± 0.23	95.82 ± 0.53	95.76 ± 0.54

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Signal	$K = 1$		$K = 2$		$K = 3$	
	Acc (%)	**F1 (%)**	**Acc (%)**	**F1 (%)**	**Acc (%)**	**F1 (%)**
A	82.57 ± 0.29	82.61 ± 0.28	83.08 ± 0.42	83.08 ± 0.46	82.88 ± 0.64	82.94 ± 0.63
V+A	97.60 ± 0.07	97.58 ± 0.08	97.87 ± 0.22	97.85 ± 0.22	97.66 ± 0.10	97.63 ± 0.10
Avg	91.73	91.70	92.24	92.22	92.12	92.11

4.6.2. Impact of fine-tuning parameters

To evaluate the impact of LoRA fine-tuning parameters on model performance, an ablation study was conducted on the IB dataset by varying the low-rank dimension r, as summarized in Table 10. As r increases from 4 to 32, the number of trainable parameters grows from approximately 5.2 million to 41.6 million, while the total model size remains around 1.97 billion parameters. The time per training epoch stays nearly constant at around 11 min, indicating minimal computational overhead due to LoRA.

Table 10. Impact of fine-tuning parameters on the IB dataset .

r	**Trainable parameters**	**Time per epoch**	**Acc (%)**	**F1 (%)**
4	5,199,952	11 min 0 s	97.87 ± 0.22	97.85 ± 0.22
8	10,399,904	11 min 9 s	96.75 ± 0.14	96.78 ± 0.14
16	20,799,808	11 min 10 s	97.30 ± 0.22	97.33 ± 0.21
32	41,599,616	11 min 22 s	96.65 ± 0.34	96.64 ± 0.35

In terms of diagnostic performance, the model achieves the highest accuracy and F1-score (97.87%/97.85%) when r =4. Increasing the rank beyond this setting does not lead to further improvements and even results in a slight decline (e.g., 96.64% at r=32). This phenomenon can be explained by the theoretical foundation of LoRA, which assumes that task-specific weight updates lie in a low-dimensional subspace and can therefore be effectively approximated using low-rank decomposition [36]. As a result, only a small number of additional parameters are sufficient to capture the necessary adaptation. Since the backbone used in this work is a relatively small 1.5B-parameter model, adopting an even smaller rank is reasonable and provides an effective balance between efficiency and accuracy.

4.7. Analysis of deployability

To assess deployability, the proposed method was compared with other LLM-based methods. All approaches employed the same backbone model, DeepSeek-R1-Distill-Qwen-1.5B. For the signal feature extraction stage, identical baseline modules were adopted to ensure fairness. The experiments were conducted on a single NVIDIA RTX 4070Ti GPU with 12 GB memory.

The results, as shown in Table 11, indicate that GPU memory usage and inference latency of the proposed method remain in the same order of magnitude as those of the baselines. Although a slight increase in resource consumption is observed, the improvement in diagnostic accuracy demonstrated in the preceding experiments justifies this trade-off. Furthermore, the proposed approach provides flexible handling of heterogeneous signal fusion tasks, which is crucial for industrial applications. For practical deployment, optimization can be further achieved through strategies such as prompt compression, mixed-precision inference, or expert pruning.

Table 11. Deployability comparison of different models.

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Model	Task type	Params (M)	Encode (TF)	Decode (GF/tok)	Total (TF/@32)	Latency (s/sample)
BearLLM	Generation	1782.26	0.281	3.105	0.380	0.0297
LLMFB-1	Generation	1777.63	0.743	3.096	0.842	0.0500
LLMFB-2	Classification	1777.62	–	–	0.604	0.0317
Proposed (M**=2,** K**=2**)	Generation	1962.18	0.428	2.938	0.522	0.0411

Note. Params (M): number of model parameters in millions. Encode (TF): encoding FLOPs per sample (in teraFLOPs). Decode (GF/tok): decoding FLOPs per token per sample (in gigaFLOPs). Total (TF/@32): estimated total FLOPs per sample assuming 32 generated tokens. Latency (s/sample): average inference latency per sample in seconds.

5. Conclusions

This paper proposes a MoE-LLM-based multisensor information fusion method for fault diagnosis. The approach employs a multisensor embedding layer to transform various types and combinations of heterogeneous sensor signals into a unified feature representation. The MoE-LLM model processes these representations effectively through an expert routing mechanism. Experimental results demonstrate that the proposed method achieves excellent diagnostic performance across multiple datasets.

Despite the promising results, several limitations remain. First, while the present work mainly considers vibration, acoustic, and current signals, extending the framework to other signal types such as temperature or image data may introduce additional challenges due to heterogeneous feature distributions and distinct sampling characteristics. Second, the adopted LLM backbone is relatively small in scale (1.5B parameters), and thus may not fully leverage the potential of large-scale pre-trained language models. Third, as the number of sensor signal types increases, the required number of expert sub-networks may grow exponentially, which could limit the model's scalability. Fourth, although the model supports flexible combinations of sensor signals, it does not explicitly address potential temporal misalignment or asynchronous sampling across signals, which may occur in real-world industrial scenarios.

Future work will focus on the following directions: (1) extending the model to new signal types and evaluating its scalability in broader industrial scenarios; (2) developing diagnostic frameworks based on larger-scale pre-trained LLMs and incorporating more efficient parameter adaptation methods; (3) designing more compact and adaptive expert selection and routing mechanisms to enhance the scalability of the MoE structure; and (4) supporting asynchronous or event-driven sensor data to improve applicability in complex real-world environments.

CRediT authorship contribution statement

Tantao Lin: Writing – original draft, Software, Methodology, Conceptualization. **Zhijun Ren:** Writing – original draft, Visualization, Investigation. **Kai Huang:** Writing – original draft, Visualization, Investigation. **Hamid Reza Karimi:** Writing – original draft, Visualization, Supervision, Investigation. **Yongsheng Zhu:** Writing – original draft, Visualization, Supervision, Investigation. **Ke Feng:** Writing – review & editing. **Jun Hong:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Recommended articles

Data availability

The authors do not have permission to share data.

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